Hints: Methods of proof – 4. Invariants, monovariants and double counting

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Look at the parity of the sum of the numbers on the board: replacing a, b by $|a-b|$ changes the sum by $2\min(a,b)$.

Hint 2. So the parity of the sum is invariant, and the last number has the parity of $1+\cdots+n = \frac{n(n+1)}{2}$. Zero is possible only if $n \equiv 0$ or $3 \pmod 4$.

Hint 3. Construction: four consecutive numbers $k, k+1, k+2, k+3$ give $1, 1$ and then 0; for $n \equiv 3$ first turn $1, 2, 3$ into $3-2=1$ and $1-1=0$. Then combine the zeros ($|0-0|=0$).

Problem 2

Hint 1. Colour the cell (i,j), $0 \leqslant i, j \leqslant 9$, with colour $(i+j) \bmod 4$. Every 1×4 piece covers each colour exactly once.

Hint 2. Count the colours on the 10×10 board: they occur $25, 26, 25, 24$ times (colours $0, 1, 2, 3$), not equally often, so no tiling.

Hint 3. General case: an $m \times n$ board is tileable by $1 \times k$ pieces iff $k \mid m$ or $k \mid n$. "If" is clear. For "only if", put the weight ω^{i+j} on the cell (i,j), $\omega = e^{2\pi i/k}$: each piece has total weight 0, while the board has weight $\left(\sum_{i<m}\omega^i\right)\left(\sum_{j<n}\omega^j\right)$, which vanishes only if $k \mid m$ or $k \mid n$.

Problem 3

Hint 1. Watch the total number of hostile pairs.

Hint 2. If the person has e enemies and f friends with $e > f$, after the switch he has f enemies and e friends, and no other pair changes. So the number of hostile pairs drops by $e - f \geqslant 1$.

Hint 3. It is a non-negative integer at most $\binom{n}{2}$ at the start, so there are at most $\binom{n}{2}$ steps.

Problem 4

Hint 1. $\sum_k \binom{n}{k}^2 = \binom{2n}{n}$: choose n people from n men and n women, and sort by the number k of men; use $\binom{n}{k} = \binom{n}{n-k}$.

Hint 2. $\sum_k k\binom{n}{k} = n2^{n-1}$: count committees with a chairman from n people, choosing the committee first or the chairman first. A second way: differentiate $(1+x)^n = \sum \binom{n}{k}x^k$ and put $x = 1$.

Hint 3. Vandermonde $\sum_k \binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r}$: an r-subset of $M \sqcup N$ ($|M| = m$, $|N| = n$) corresponds bijectively to the pair (its part in M, its part in N); sort by the size k of the first part.

Problem 5

Hint 1. Let the design have v points, b blocks of size $k < v$, every point in r blocks and every pair of points in $\lambda \geqslant 1$ blocks. Counting pairs (x, A) with $x \in A$ gives $bk = vr$; counting triples (x, y, A), $x \neq y$ in A, with x fixed gives $r(k-1) = \lambda(v-1)$.

Hint 2. Fisher's inequality $b \geqslant v$ is equivalent to $r \geqslant k$. Fix a block B and let $m_C = |B \cap C|$ for the other $b-1$ blocks. Double count again: $\sum_C m_C = k(r-1)$ and $\sum_C m_C(m_C-1) = k(k-1)(\lambda-1)$.

Hint 3. Cauchy–Schwarz $(\sum m_C)^2 \leqslant (b-1)\sum m_C^2$; after substituting λ and b from Hint 1 it simplifies to $\frac{r(r-k)(v-k)^2}{v-1} \geqslant 0$, so $r \geqslant k$ and $b \geqslant v$. (The usual alternative is the rank of $NN^T = (r-\lambda)I + \lambda J$.)

Problem 6

Hint 1. Trace and characteristic polynomial are invariant under similarity. Over $\mathbb{C}$ (or the algebraic closure) a nilpotent N is similar to a strictly upper triangular matrix, since its only eigenvalue is 0.

Hint 2. Hence $\operatorname{tr} N = 0$ (the sum of the eigenvalues with multiplicity, all zero). For the commutator use $\operatorname{tr}(AB) = \operatorname{tr}(BA)$: $\operatorname{tr}(AB - BA) = 0$, while $\operatorname{tr} E = n \neq 0$ in characteristic 0.

Hint 3. (In characteristic p with $p \mid n$ the trace argument fails, and indeed $AB - BA = E$ is then possible, e.g. for $n = p$ with differentiation and multiplication by x on $\mathbb{F}_p[x]/(x^p)$.)

Problem 7

Hint 1. First check that swapping two adjacent entries changes the number of inversions by exactly ± 1.

Hint 2. A transposition of the entries in positions $i < j$ is a composition of $2(j-i)-1$ adjacent swaps, an odd number; so the parity of the number of inversions flips. Hence $\operatorname{sgn}\colon S_n \to \{\pm 1\}$ is a well-defined homomorphism.

Hint 3. For $n \geqslant 2$ it is surjective, so its kernel A_n has index 2. Application: the 15-puzzle (class work, problem 1), or the Rubik's cube fact that a single swap of two edge pieces is impossible.

Problem 8

Hint 1. Extremal proof: if the set has at least two points, take two points A, B at the minimal distance. What about A and the midpoint of AB?

Hint 2. The midpoint M is in the set and $|AM| = |AB|/2 < |AB|$: contradiction. Monovariant (descent) proof: starting from $A \neq B$, the points $A + (B-A)/2^k$, $k = 0, 1, 2, \ldots$, all belong to the set and are pairwise distinct, contradicting finiteness.

Hint 3. Lattice comparison: $\mathbb{Z}^2$ is not midpoint-closed, but among any five lattice points two have the same parity pattern of coordinates, so their midpoint is a lattice point; there the obstruction is a residue (pigeonhole on $\mathbb{Z}^2/2\mathbb{Z}^2$) instead of a size.

Problem 9

Hint 1. If n has at least two digits, then $S(n) < n$: compare $\sum d_i 10^i$ with $\sum d_i$. So the value strictly decreases until it is a single digit.

Hint 2. Since $10^i \equiv 1 \pmod 9$, $n \equiv S(n) \pmod 9$: the residue modulo 9 is invariant.

Hint 3. The process ends at the unique digit $1, \ldots, 9$ congruent to n modulo 9: the digital root of $n \geqslant 1$ is $1 + ((n-1) \bmod 9)$.

Problem 10

Hint 1. Take a connected plane graph. If it contains a cycle, delete one edge of the cycle. What happens to V, E and F?

Hint 2. The cycle separates the plane (Jordan curve theorem), so the edge borders two different faces, which merge: E and F both drop by 1 and $V - E + F$ is unchanged. E is a decreasing monovariant, so the process ends in a tree, where $V - E + F = V - (V - 1) + 1 = 2$.

Hint 3. K_5: if it were planar, $F = 2 - 5 + 10 = 7$. Count edge–face incidences: each face is bounded by at least 3 edges, each edge borders at most 2 faces, so $3F \leqslant 2E = 20$, i.e. $F \leqslant 6$. Contradiction.